

Element	Transfer Function	Bode Feature	
proportional	$G(s) = K$	$L(\omega) = 20 \log_{10} K$,	$\varphi = 0^\circ$
integrator	$G(s) = 1/s$	slope -20 dB/dec,	$\varphi = -90^\circ$
differentiator	$G(s) = s$	slope $+20$ dB/dec,	$\varphi = +90^\circ$
inertia	$G(s) = 1/(Ts + 1)$	break at $\omega = 1/T$, then	-20 dB/dec
lead / first-order diff.	$G(s) = 1 + \tau s$	break at $\omega = 1/\tau$, then	$+20$ dB/dec
oscillatory	$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$	high-frequency -40 dB/dec,	slope phase $0^\circ \rightarrow -180^\circ$